# Control Systems
# CSE481

## Project

## Under supervision of

**Dr. Michael Ibrahim**
**Eng. Mina**

## Prepared by:

**Ahmed Sherif 23P0414**

**Ahmed Belal 23P0007**

**Marina Amgad 23P0330**

Code Link:
https://drive.google.com/file/d/14mYgUx3AXeu3ZeJT_kwpGPQxnmUtJ7qP/view?usp=drive_link

# Contents


..... 1

..... 1

Abstract ..... 3

1) Mathematical Model for the Traction Drive Motor with a Capstan Roller and Linear Slide ..... 4

1. Electrical Equation ..... 4
2. Motor torque Generation ..... 4
3. Linear Slide Dynamics ..... 4
4. Traction Drive Coupling ..... 5
5. Mechanical Subsystem ..... 5
6. Complete System Equations ..... 6
   a) Electrical equation ..... 6
   b) Torque equation ..... 6
   c) Mechanical equation ..... 6
   d) Kinematic constraint ..... 6
   e) Position-velocity relationship ..... 6
7. Transfer Functions ..... 6

2) State Variable Model ..... 7

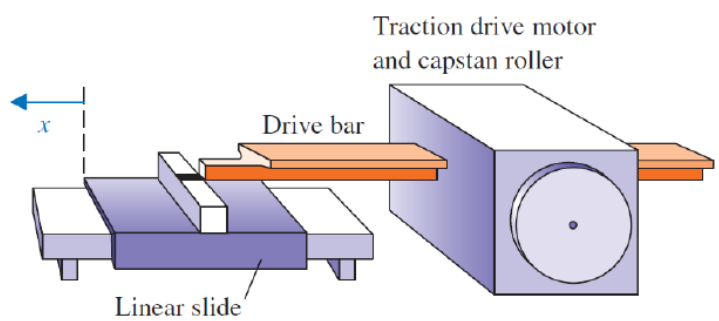
1. Define the system states, input, and outputs ..... 7
2. Write the differential equations of motion ..... 7
3. Write in state-space (matrix) form ..... 9
4. Substitute numerical values ..... 10
5. Numerical matrices ..... 10
6. Final State-Space Representation ..... 10

CODE OF PROJECT ..... 11

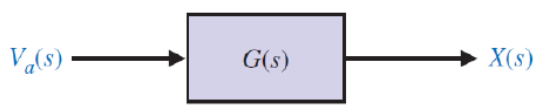
Hand ANALYSIS ..... 18

# Abstract

It is required to design a system which can be used to accurately position a table for a machine as shown below.




(a)




A traction-drive motor with a capstan roller possesses several desirable characteristics compared to the more popular ball screw mechanism. The traction drive exhibits low friction and no backlash. However, it is susceptible to disturbances. The system uses a DC armature-controlled motor with a capstan roller attached to the shaft. The drive bar moves the linear slide-table. The slide uses air bearing, so its friction is negligible.

A model of the traction drive is developed for the parameters given in Table below.

| Typical Parameters for the Armature-Controlled DC Motor and the Capstan and Slide | | |
|---|---|---|
| $M_s$ | Mass of slide | 5.693 kg |
| $M_b$ | Mass of drive bar | 6.96 kg |
| $J_m$ | Inertia of roller, shaft, motor and tachometer | $10.91 \times 10^{-3}$ kg m$^2$ |
| $r$ | Roller radius | $31.75 \times 10^{-3}$ m |
| $b_m$ | Motor damping | 0.268 N ms/rad |
| $K_m$ | Torque constant | 0.8379 N m/amp |
| $K_b$ | Back emf constant | 0.838 V s/rad |
| $R_m$ | Motor resistance | 1.36 Ω |
| $L_m$ | Motor inductance | 3.6 mH |

# 1) Mathematical Model for the Traction Drive Motor with a Capstan Roller and Linear Slide

The system has one input, which is the voltage $V_a(s)$ applied to the motor. The variable $i_a(t)$ or $I_a(s)$ is the armature current. The system has three outputs which are:

- The angular speed of the capstan roller, $\Omega(s)$
- The angular position of the capstan roller, $\theta(s)$
- The displacement of the linear slide, $X(s)$

**These are the complete system equations:**

## 1. Electrical Equation

The armature circuit equation is

$$v_a(t) = R_m i_a(t) + L_m \frac{di_a(t)}{dt} + e_b(t)$$

$v_a$ = applied armature voltage

$e_b(t)$ = back EMF

Since the back EMF is proportional to the angular velocity; therefore,

$$e_b(t) = K_b \omega(t)$$

$\omega(t)$ = angular speed of motor shaft

Applying Laplace Transform,

$$V_a(s) = (R_m + L_m s) I_a(s) + K_b \Omega(s)$$

## 2. Motor torque Generation

The electromagnetic torque produced by the motor is represented by

$$T_m(t) = K_m i_a(t)$$

Applying Laplace Transform,

$$T_m(s) = K_m I_a(S)$$

## 3. Linear Slide Dynamics

With negligible friction (air bearing), the force balance on the slide:

$$F(t) = M\frac{dv(t)}{dt} = M\frac{d^2x(t)}{dt^2}$$

$M = M_s + M_b$ is total mass of drive bar and the linear slide and table

$F$ = force applied by. the capstan roller

The force-and-torque relationship can be expressed by

$$F(t) = \frac{T_L(t)}{r}$$

where $T_L(t)$ is the load torque from the linear slide.

Applying the Laplace Transform,

$$F(s) = Ms^2X(s) = \frac{T_L(s)}{r}$$

Therefore,

$$T_L(s) = rMs^2X(s)$$

## 4. Traction Drive Coupling

The capstan roller with radius $r$drives the linear slide. For no-slip condition:

$$v(t) = \frac{dx(t)}{dt} = r\omega(t)$$

$v(t)$ is the linear velocity of the slide.

Applying Laplace Transform,

$$sX(s) = r\Omega(s)$$

## 5. Mechanical Subsystem

The equation of motion for the rotating components (motor rotor + capstan roller):

$$T_m(t) - T_L(t) = J_m\frac{d\omega(t)}{dt} + b_m\omega(t)$$

Applying Laplace Transform and substituting the electromagnetic and the load torques with our variables extracted from the equations above,

$$K_mI_a(s) - rMs^2X(s) = (J_ms + b_m)\Omega(s)$$

## 6. Complete System Equations

Combining all subsystems, we come up with these equations.

### a) Electrical equation

$$I_a(s) = \frac{V_a(s) - K_b\Omega(s)}{R_a + L_a s}$$

### b) Torque equation

$$T_m = K_m I_a(s)$$

### c) Mechanical equation

$$K_m I_a(s) - rMs^2X(s) = (J_m s + b_m)\Omega(s)$$

### d) Kinematic constraint

$$X(s) = \frac{r\Omega(s)}{s}$$

### e) Position-velocity relationship

$$\theta(s) = \frac{\Omega(s)}{s}$$

## 7. Transfer Functions

### Angular Speed

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_m}{L_m J_m s^2 + (L_a b_m + R_m J_m + L_m M r^2)s + (R_m b_m + K_b K_m + R_m M r^2)}$$

### Angular Position

$$\frac{\theta(s)}{v_a(s)} = \frac{K_m}{s[L_m J_m s^2 + (L_m b_m + R_m J_m + L_m M r^2)s + (R_m b_m + K_b K_m + R_m M r^2)]}$$

### Linear Displacement

$$\frac{x(s)}{v_a(s)} = \frac{rK_m}{s[L_m J_m s^2 + (L_m bm + R_m J_m + L_m M r^2)s + (R_m b_m + K_b K_m + R_m M r^2)]}$$

## 2) State Variable Model

### 1. Define the system states, input, and outputs

Let the **state variables** be:

$$x_1 = i_a(t)(\text{armature current [A]})$$
$$x_2 = \omega(t)(\text{angular speed of the roller [rad/s]})$$
$$x_3 = x(t)(\text{linear displacement of the slide [m]})$$

The **system input** is:

$$u(t) = V_a(t)(\text{armature voltage [V]})$$

The **system outputs** are:

$$y_1 = \Omega(t) = \omega(t), y_2 = \Theta(t) = \frac{x(t)}{r}, y_3 = X(t) = x(t).$$

### 2. Write the differential equations of motion

*(a) Electrical equation (KVL)*

$$V_a(t) = R_m i_a(t) + L_m \frac{di_a(t)}{dt} + K_b \omega(t)$$

Rearrange for $i_a(t)$:

$$\boxed{\dot{i}_a = \frac{1}{L_m}\left(V_a - R_m i_a - K_b \omega\right)}$$

### *(b) Mechanical rotational equation*

$$K_m i_a(t) = J_m \frac{d\omega}{dt} + b_m \omega(t) + T_L(t)$$

The load torque due to the slide mass is:

$$T_L = rF = rM\ddot{x} = Mr^2\dot{\omega}$$

Substitute into the above:

$$K_m i_a = (J_m + Mr^2)\dot{\omega} + b_m \omega$$

$$\boxed{\dot{\omega} = \frac{1}{J_m + Mr^2}(K_m i_a - b_m \omega)}$$

### *(c) Kinematic relation*

$$\boxed{\dot{x} = r\,\omega}$$

## 3. Write in state-space (matrix) form

Let the state vector be:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} i_a \\ \omega \\ x \end{bmatrix}$$

The state equations (from 1–3) are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\dfrac{R_m}{L_m} & -\dfrac{K_b}{L_m} & 0 \\ \dfrac{K_m}{J_m + Mr^2} & -\dfrac{b_m}{J_m + Mr^2} & 0 \\ 0 & r & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \dfrac{1}{L_m} \\ 0 \\ 0 \end{bmatrix} V_a$$

or compactly:

$$\boxed{\dot{\mathbf{x}} = A\mathbf{x} + Bu}$$

The output equations are:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & \dfrac{1}{r} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} V_a$$

or

$$\boxed{y = C\mathbf{x} + Du}$$

## 4. Substitute numerical values

Given parameters:

- $M_s = 5.693 \text{ kg}$
- $M_b = 6.96 \text{ kg}$
- $M = M_s + M_b = 12.653 \text{ kg}$
- $J_m = 10.91 \times 10^{-3} = 0.01091 \text{ kg} \cdot \text{m}^2$
- $r = 31.75 \times 10^{-3} = 0.03175 \text{ m}$
- $b_m = 0.268 \text{ N} \cdot \text{m} \cdot \text{s/rad}$
- $K_m = 0.8379 \text{ N} \cdot \text{m/A}$
- $K_b = 0.838 \text{ V} \cdot \text{s/rad}$
- $R_m = 1.36 \, \Omega$
- $L_m = 3.6 \times 10^{-3} = 0.0036 \text{ H}$

## 5. Numerical matrices

$$A = \begin{bmatrix} -377.78 & -232.78 & 0 \\ 35.4067 & -11.3247 & 0 \\ 0 & 0.03175 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 277.78 \\ 0 \\ 0 \end{bmatrix},$$

$$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 31.4961 \\ 0 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

## 6. Final State-Space Representation

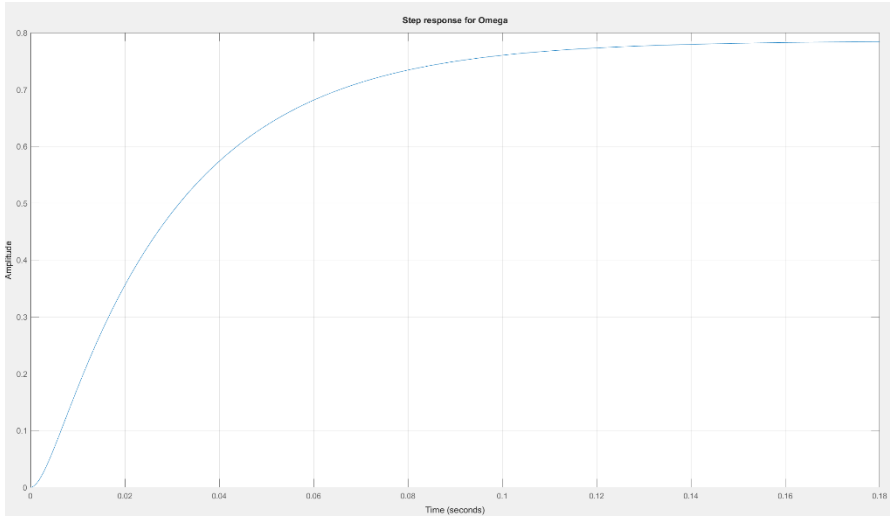
$$\boxed{\begin{aligned} \dot{\mathbf{x}} &= A\mathbf{x} + Bu \\ \mathbf{y} &= C\mathbf{x} + Du \end{aligned}}$$

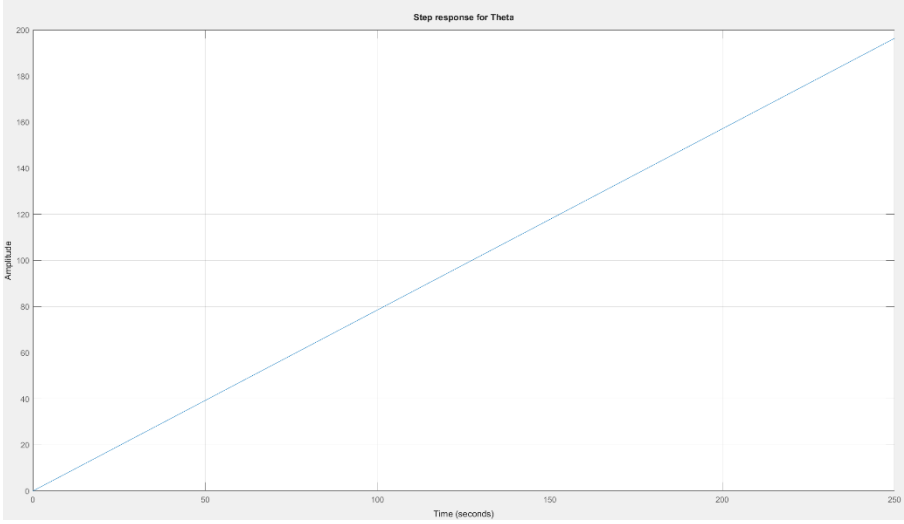
# CODE OF PROJECT

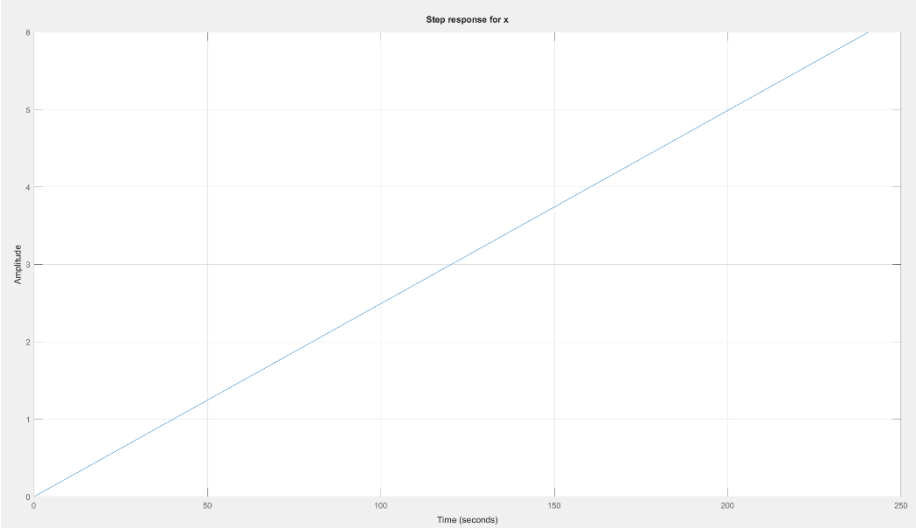
```
clear; clc; close all;

%%System Parameters
Ms = 5.693; %mass of slide
Mb = 6.96; %mass of drive bar
M = Ms + Mb;
Jm = 10.91e-3; %inertia
r = 31.75e-3; %roller radius
bm = 0.268; %motor damping
Km = 0.8379; %torque constant
Kb = 0.838; %back EMF constant
Rm = 1.36; % motor resistance
Lm = 3.6e-3; %motor inductance

A = [-Rm/Lm -Kb/Lm 0; Km/(Jm+M*(r^2)) -bm/(Jm+M*(r^2)) 0; 0 r 0];
B = [1/Lm; 0; 0];
C = [0 1 0; 0 0 1/r; 0 0 1];
D = [0; 0; 0];

sys = ss(A, B, C, D); % Create state-space system representation
tf_omega = tf(sys(1)); %Output1: omega(t)
tf_theta = tf(sys(2)); %Output2: theta(t)
tf_x = tf(sys(3)); % Output3: x(t)

%Step Response of each output
% omega(t)
figure
step(sys(1))
title('Step response for Omega')
grid on
% theta(t)
figure
step(sys(2))
title('Step response for Theta')
grid on
% x(t)
figure
step(sys(3))
title('Step response for x')
grid on
```

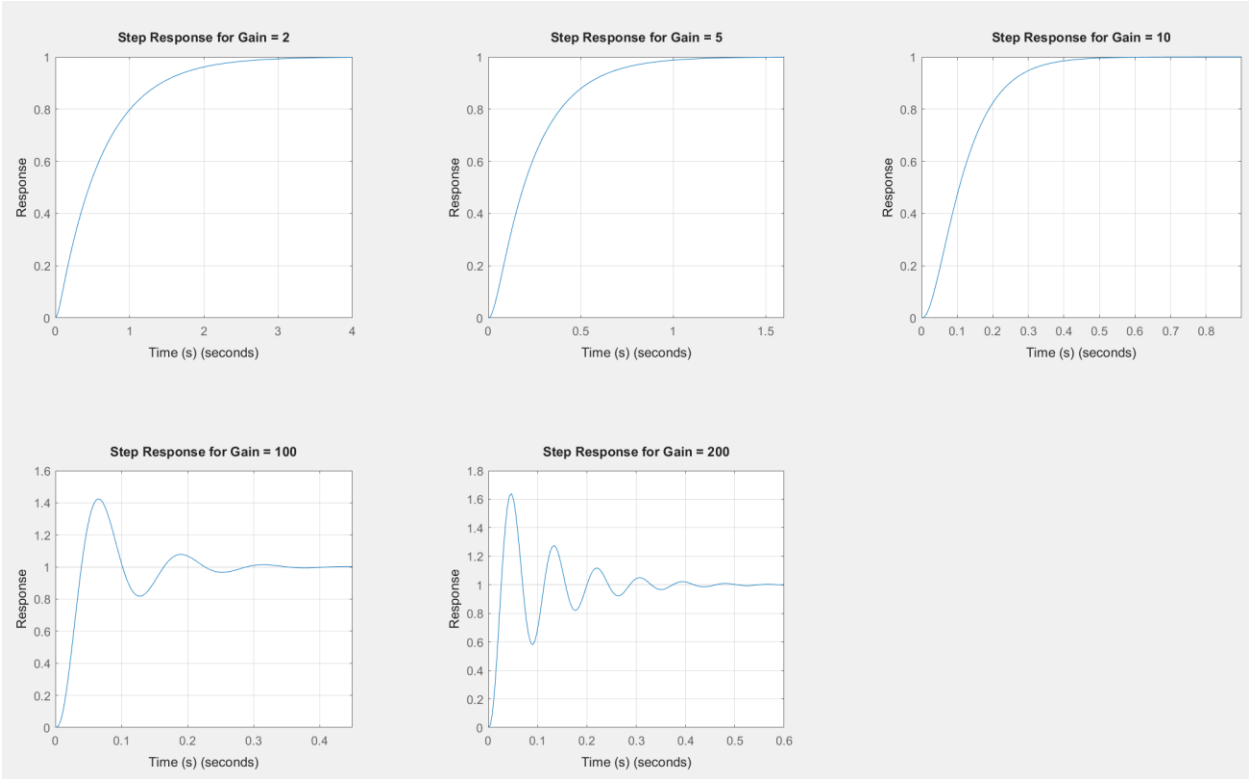
```
%%%%%%%%%%%%%%%%%%%%%%%%%%% Point No.5 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
s = tf('s'); %define the laplace variable
Gc = [2, 5, 10, 100, 200];
G1 = tf_omega;
integrator = 1/s;

%*Step Input*%
% Create figure for step responses
figure;
% Loop through each gain value in Gc and plot the step response
for i = 1:length(Gc)
    L = integrator* G1* Gc(i); %open loop
    T = feedback(L, 1);
    subplot(2, 3, i);
    step(T); % Step response for the scaled system
    title(['Step Response for Gain = ', num2str(Gc(i))]);
    xlabel('Time (s)');
    ylabel('Response');
    grid on;
end

%*Ramp Input*%
t = 0:0.01:10;
ramp = t; % simple ramp input
% Create figure for ramp responses
figure;
% Loop through each gain value in Gc and plot the ramp response
for i = 1:length(Gc)
    L = integrator* G1* Gc(i); % open loop
    T = feedback(L, 1);
    subplot(2, 3, i);
    lsim(T, ramp, t); % Ramp response for the scaled system
    title(['Ramp Response for Gain = ', num2str(Gc(i))]);
    xlabel('Time (s)');
    ylabel('Response');
    grid on;
end

%%%%%%%%%%%%%%Point 7%%%%%%%%%%%%%%%%%%
%%%%% Without disturbance %%%%%
Ku = 495.3260803; %Ku obtained from hand analysis
Tu = 0.05;

Kp = 0.6 * Ku;
Ki = 1.2 * Ku / Tu;
Kd = 0.6 * Ku * Tu / 8;

Gc_pid = Kp + Ki/s +s*Kd;
% Create a closed-loop system with the PID controller
L_pid = integrator* G1* Gc_pid; % open loop with PID
T_pid = feedback(L_pid, 1); % closed-loop system with PID
```

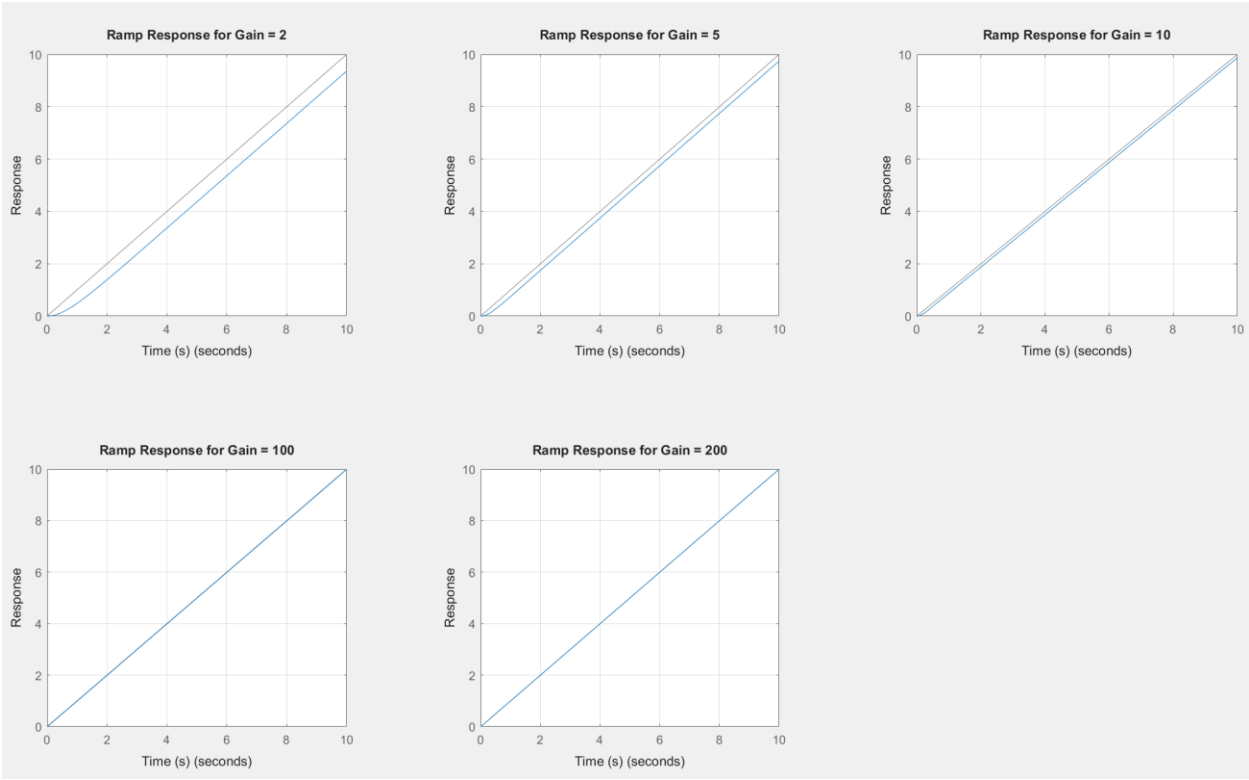
```
figure
step(T_pid)
title('Step response for Ziegler-Nichols')
grid on;

% Get the step response data for the PID controlled system
info_pid = stepinfo(T_pid);
Ts_2percent_pid = info_pid.SettlingTime;
Overshoot_percent_pid = info_pid.Overshoot;
fprintf('2%% settling time = %f\n', Ts_2percent_pid)
fprintf('Overshoot percentage = %f\n', Overshoot_percent_pid)


%%%%% System's step Response with disturbance %%%%%
T_pid_dist = feedback(-G1/s, -Gc_pid);
figure
step(T_pid_dist)
title('Response with step disturbance (ZN)')
grid on;

%%%%%%%%%%%%%%%Point 8%%%%%%%%%%%%%%%%%%%
%%%%% Without disturbance %%%%%
KP = 5;
KI = 0.01;
KD = 0.5;
Gc_pid_tuned = KP + KI/s + s*KD; % Define the PID controller with new gains
L_pid_tuned = integrator * G1 * Gc_pid_tuned; % Open loop with tuned PID
T_pid_tuned = feedback(L_pid_tuned, 1); % Closed-loop system with tuned PID

figure
step(T_pid_tuned)
title('Step response for my colleague')
grid on;

% Get the step response data for the PID controlled system
info_pid_tuned = stepinfo(T_pid_tuned);
Ts_2percent_pid_tuned = info_pid_tuned.SettlingTime;
Overshoot_percent_pid_tuned = info_pid_tuned.Overshoot;
fprintf('My colleague has 2%% settling time = %f\n', Ts_2percent_pid_tuned)
fprintf('My colleague has Overshoot percentage = %f\n', Overshoot_percent_pid_tuned)

%%%%% System's step Response with disturbance %%%%%
T_pid_tuned_dist = feedback(-G1/s, -Gc_pid_tuned);
figure
step(T_pid_tuned_dist)
title('Response with step disturbance (colleague)')
grid on;
```

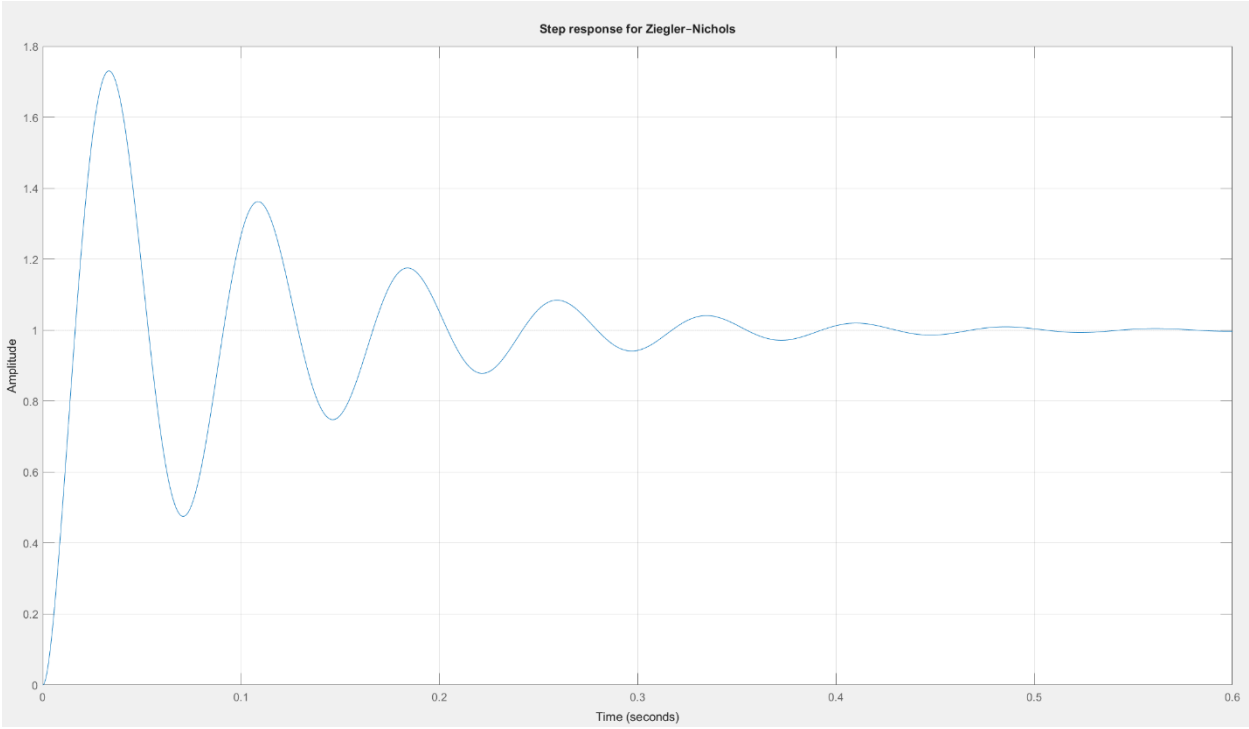

Step response for Omega
Amplitude
Time (seconds)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18

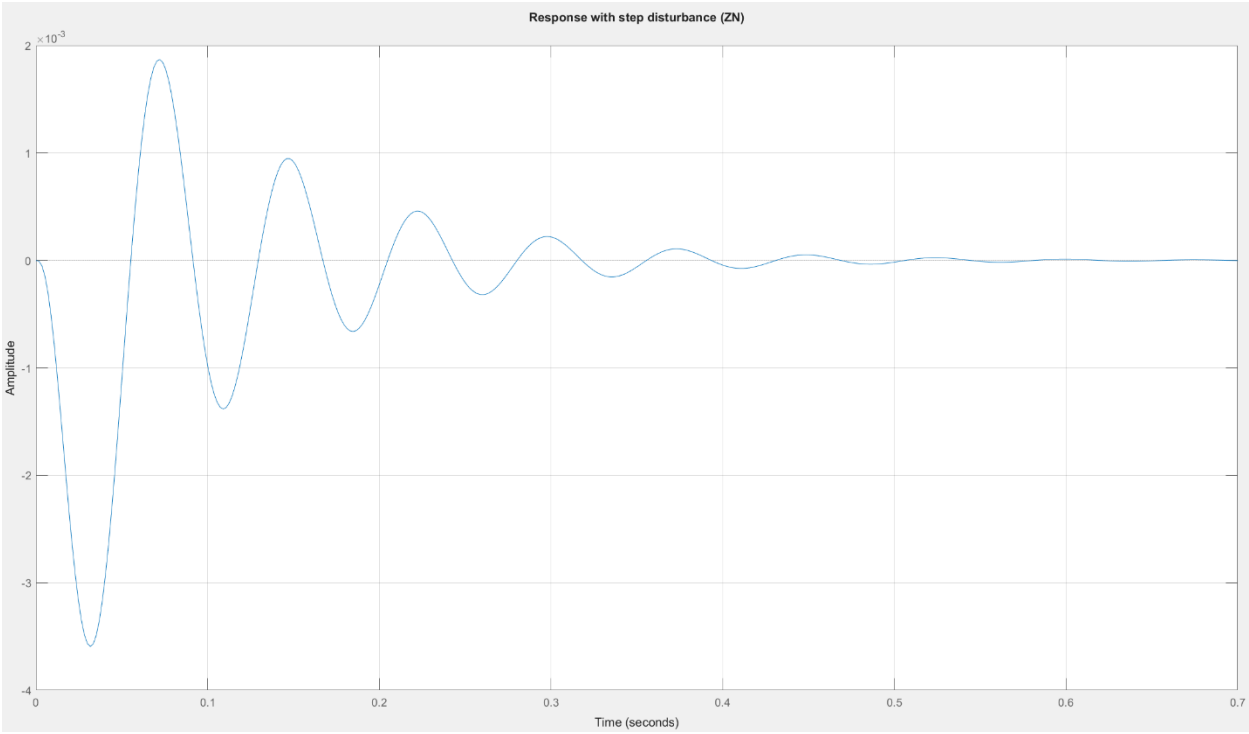

Step response for Theta
Amplitude
Time (seconds)
0
20
40
60
80
100
120
140
160
180
200
50
150
250

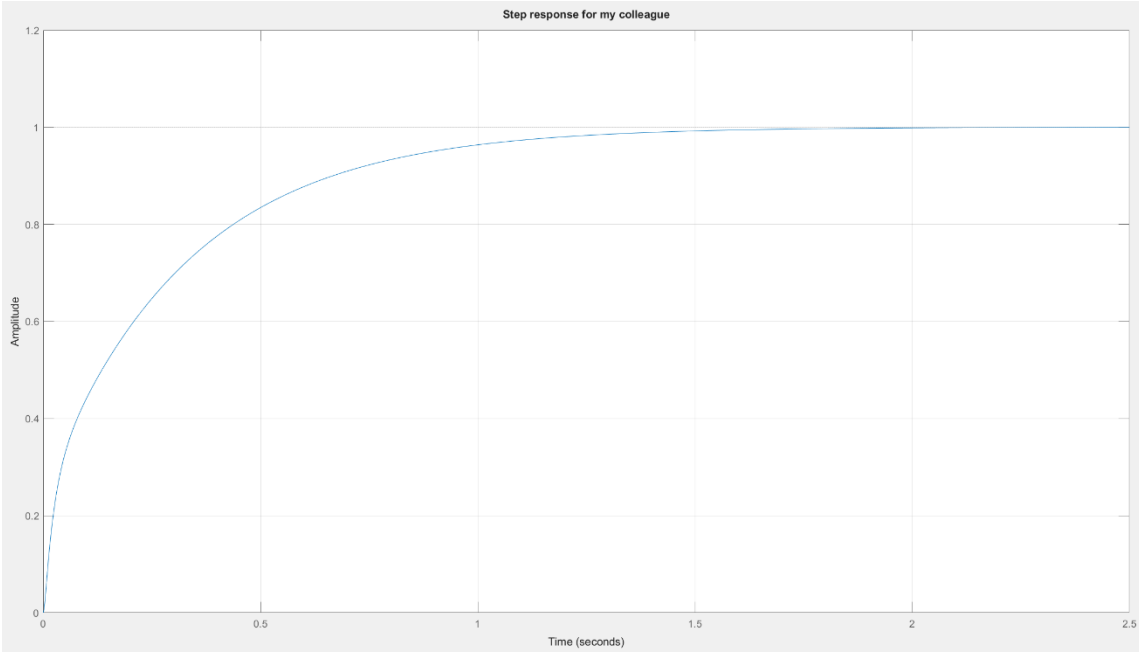

Step response for x
Amplitude
Time (seconds)
0
1
2
3
4
5
6
50
100
150
200
250

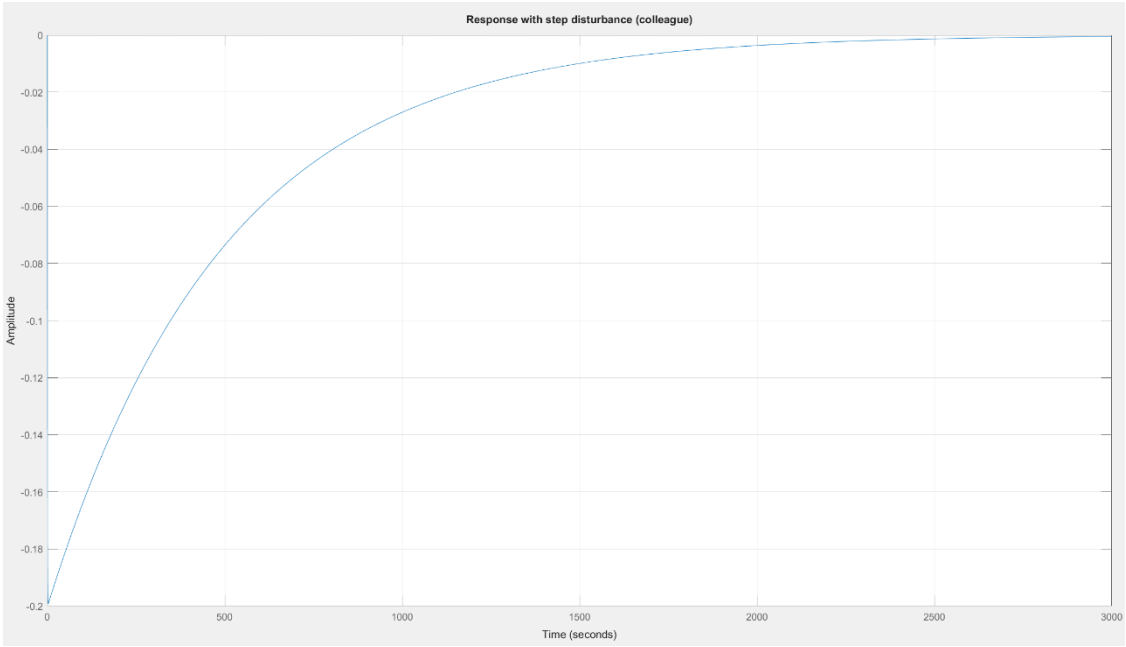

Step Response for Gain = 2
Step Response for Gain = 5
Step Response for Gain = 10
Step Response for Gain = 100
Step Response for Gain = 200
Response
Time (s) (seconds)


Ramp Response for Gain = 2
Ramp Response for Gain = 5
Ramp Response for Gain = 10
Ramp Response for Gain = 100
Ramp Response for Gain = 200
Response
Time (s) (seconds)


Step response for Ziegler-Nichols
Amplitude
Time (seconds)
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
0.6


Response with step disturbance (ZN)
×10^-3
Amplitude
Time (seconds)
2
1
0
-1
-2
-3
-4
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7




```
2% settling time = 0.382201
Overshoot percentage = 73.069613
My colleague has 2% settling time = 1.192483
My colleague has Overshoot percentage = 0.037702
```

## Hand ANALYSIS

Point (6)

$$T(s) = \frac{G_1 G_c}{s + G_1 G_c} = \frac{G_{1u} G_c}{G_{1d} s + G_{1u} G_c}$$

$$q(s) = s^3 + 389.1 s^2 + (1.252 \times 10^4) s + 9835 k_a$$

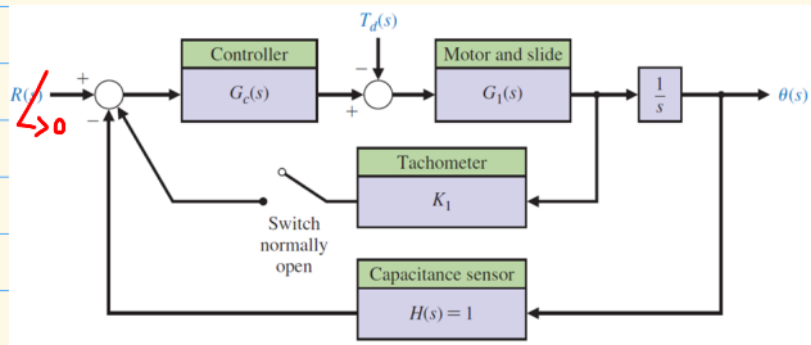
| | | |
|---|---|---|
| $s^3$ | $1$ | $1.252 \times 10^4$ |
| $s^2$ | $389.1$ | $9835 k_a$ |
| $s^1$ | $b_1$ | |
| $s^0$ | $c_1$ | |

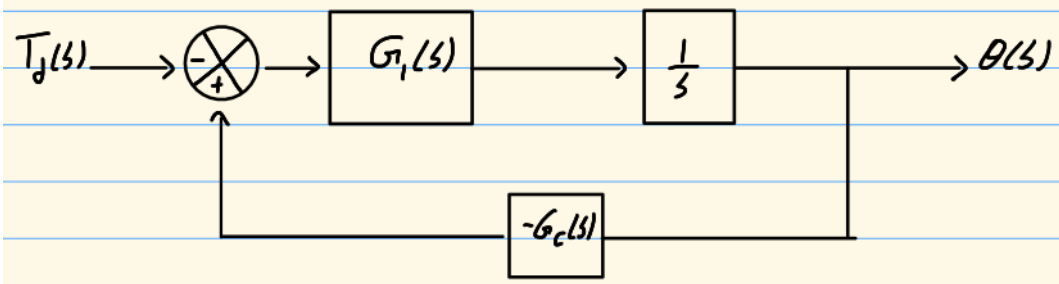
$$b_1 = \frac{1}{389.1}(4871532 - 9835 k_a)$$

$$c_1 = 9835 k_a$$

- $b_1 > 0$
  $k_a < 495.326$
- $c_1 > 0$
  $k_a > 0$

$$K_u = 495.326$$




$$\frac{\theta(s)}{T_d(s)} = \frac{-G_1(s)\cdot 1/s}{1-(G_1(s)\cdot 1/s)(-G(s))} = \frac{-G_1(s)}{s+G_1(s)G(s)}$$

**4) Do you observe any steady state error in any of the outputs?**

For the angular speed $\Omega(t)$, the step response reaches a finite constant value, therefore no steady-state error is observed. However, for the angular position $\theta(t)$and the linear position $x(t)$, the responses increase without bound and tend to infinity. As a result, a steady state is not reached for these outputs, and hence the steady-state error is undefined.

**8) Which design behaves better? Why?**

- Settling Time: Ziegler–Nichols system is better because it shows a faster response.
- Overshoot: The colleague's system is better as overshot is approximately zero.